



TECHNICAL DOCUMENTATION

STP AND ETHERCHANNEL NETWORK

Cisco Packet Tracer project

ARCHITECTURE	Core / Access
REDUNDANCY	LACP EtherChannel
SEGMENTATION	VLANs 10, 20 and 30
SECURITY	SSH, AAA, PortFast and BPDU Guard

Document control

Field	Value
Document title	Technical Documentation - STP and EtherChannel Network
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Platform	Cisco Packet Tracer
Devices in scope	CORESW1, CORE2, ACCESS1, ACCESS2
Sources	Lab04_STPthChannel.pkt; SWCORE 1.txt; SWCORE2.txt; ACCESS 1(1).txt; ACCESS 2(1).txt

SCOPE AND RELIABILITY

Topology, ports, VLAN assignments, STP roles, trunks and EtherChannel state were derived directly from four running-config/show exports. Exact switch models and endpoint IP settings are not present in the text exports and are therefore not presented as confirmed facts.

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1. Introduction and project description

The project is a layered Layer 2 network built in Cisco Packet Tracer. It consists of two core switches and two access switches. The core devices are connected by two GigabitEthernet links combined into a LACP EtherChannel, while each access switch is connected to its corresponding core switch through a two-link FastEthernet EtherChannel.

User traffic is segmented into three VLANs: VLAN 10 for Management, VLAN 20 for Sales and VLAN 30 for IT. All inter-switch links operate as static IEEE 802.1Q trunks and carry only VLANs 10, 20 and 30. PVST maintains a separate STP instance for every VLAN.

SOLUTION SUMMARY

The project demonstrates VLAN segmentation, LACP link aggregation, PVST root-bridge election, PortFast and BPDU Guard protection on access ports, and safer management through SSHv2 with local AAA. It is a Layer 2 lab: no SVI addressing, inter-VLAN routing or default gateways are configured.

2. Project objectives and requirements

The main objective is to build a stable and segmented switched network that uses link aggregation and STP to control redundant physical links.

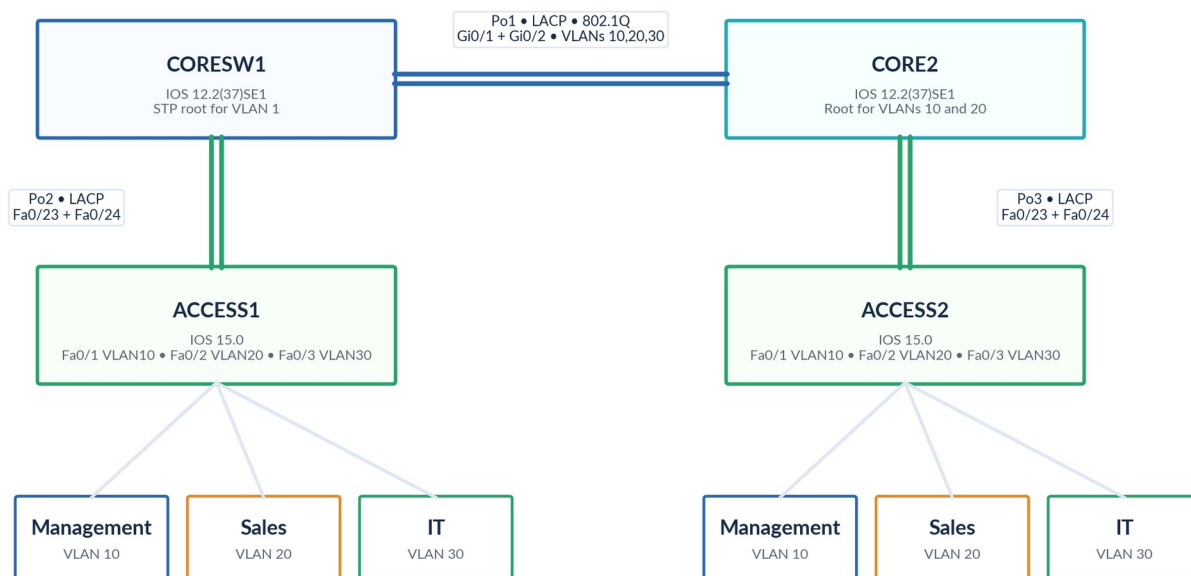
- Separate Management, Sales and IT users into individual broadcast domains.
- Carry VLANs 10, 20 and 30 between switches over 802.1Q trunks.
- Combine parallel links by using LACP EtherChannel.
- Prevent loops and define the STP root bridge deterministically.
- Accelerate endpoint-port activation by using PortFast.
- Protect access ports against unwanted BPDUs by using BPDU Guard.
- Restrict remote management to SSH and local AAA authentication.
- Administratively disable unused FastEthernet ports.

3. Network architecture and topology

The topology follows a core/access design. CORESW1 and CORE2 form the central layer and are connected by Port-channel1 over interfaces Gi0/1 and Gi0/2. ACCESS1 uses Port-channel2 toward CORESW1, while ACCESS2 uses Port-channel3 toward CORE2; both channels use Fa0/23 and Fa0/24.

Reconstructed physical and logical topology

Links, ports and VLANs are derived from running-config and show outputs.



Note: each access switch is connected to only one core switch; link redundancy exists, but full device-level redundancy does not.

Figure 1 - Reconstructed physical and logical network topology

IMPORTANT REDUNDANCY NOTE

EtherChannel protects against the failure of one physical channel member. However, each access switch has an uplink to only one core device. Failure of the entire core switch or port-channel isolates the associated access segment. Full device-level redundancy would require dual-homing each access switch to both core devices.

4. Devices used

Label	Type/model	IOS	Interfaces	Main functions
CORESW1	Core Layer 2 switch; exact model not confirmed	IOS 12.2(37)SE1	24× FastEthernet + 2× GigabitEthernet	Po1 to CORE2; Po2 to ACCESS1; STP root for VLAN 1
CORE2	Core Layer 2 switch; exact model not confirmed	IOS 12.2(37)SE1	24× FastEthernet + 2× GigabitEthernet	Po1 to CORESW1; Po3 to ACCESS2; root for VLANs 10/20 in the snapshot
ACCESS1	Access Layer 2 switch; exact model not confirmed	IOS 15.0	24× FastEthernet + 2× GigabitEthernet	Fa0/1-3 endpoints; Po2 uplink; PortFast/BPDU Guard
ACCESS2	Access Layer 2 switch; exact model not confirmed	IOS 15.0	24× FastEthernet + 2× GigabitEthernet	Fa0/1-3 endpoints; Po3 uplink; PortFast/BPDU Guard
Endpoints	PC/host devices in the Packet Tracer project	Not applicable	Ethernet access	At least one connection for VLANs 10, 20 and 30 on each access switch

4.1. Active and reserved ports

Ports	Mode	Purpose	Control
ACCESS1 / ACCESS2 Fa0/1	Access VLAN 10	Management host	PortFast + BPDU Guard
ACCESS1 / ACCESS2 Fa0/2	Access VLAN 20	Sales host	PortFast + BPDU Guard
ACCESS1 / ACCESS2 Fa0/3	Access VLAN 30	IT host	PortFast + BPDU Guard
ACCESS1 / ACCESS2 Fa0/4-22	Shutdown	Unused	Administratively disabled
Core Fa0/1-22	Shutdown	Unused	Administratively disabled
Access Gi0/1-2	Default state	No description/purpose	Recommendation: shut down if not required

5. Technologies and protocols

Technology/protocol	Role in the project
VLAN segmentation	Separates Management, Sales and IT traffic into VLANs 10, 20 and 30.
IEEE 802.1Q trunking	Carries multiple VLANs over Po1, Po2 and Po3; DTP is disabled with nonegotiate.
LACP EtherChannel	Active LACP mode aggregates two physical links into one logical port-channel.
PVST	Maintains a separate STP instance per VLAN and elects the root from priority + MAC values.
PortFast	Moves an access port directly to forwarding when an endpoint is connected.

Technology/protocol	Role in the project
BPDUGuard	Err-disables an access port if it receives a BPDU and detects a possible unauthorized switch.
SSHv2 + local AAA	VTY lines accept only SSH and use the local user database.
Administrative shutdown	Most unused FastEthernet ports are disabled to improve security and configuration hygiene.

6. Addressing table

The supplied configurations contain no active IPv4 address on any switch. Interface Vlan1 on all four devices explicitly has no IP address and is administratively shut down. VLANs 10, 20 and 30 are used only at Layer 2; SVI interfaces, default gateways and inter-VLAN routing are not configured.

Device/segment	Interface	IPv4 address	Status	Note
CORESW1	Vlan1	Not configured	shutdown	No IP-based management access
CORE2	Vlan1	Not configured	shutdown	No IP-based management access
ACCESS1	Vlan1	Not configured	shutdown	No IP-based management access
ACCESS2	Vlan1	Not configured	shutdown	No IP-based management access
VLAN 10	L2 segment	Subnet/gateway not defined	Active on trunk/access ports	Management segment; no L3 function
VLAN 20	L2 segment	Subnet/gateway not defined	Active on trunk/access ports	Sales segment; no L3 function
VLAN 30	L2 segment	Subnet/gateway not defined	Active on trunk/access ports	IT segment; no L3 function
Endpoints	NIC	Not confirmed in the text exports	Not confirmed	Addresses are not invented in this document

CONSEQUENCE

SSHv2 and AAA are configured, but without a management IP address and an appropriate Layer 3 path, remote SSH access is not operational. Management is currently possible through the local console, unless the .pkt project contains additional state not represented in the exports.

7. VLAN plan

VLAN	Name	Purpose	Access assignment	Transport
1	Native / default	No user purpose	Native VLAN on trunks; Vlan1 SVI shut down	Not in the explicit allowed list 10,20,30
10	Management	Management users/workstations	ACCESS1 Fa0/1; ACCESS2 Fa0/1	Po1, Po2 and Po3 trunks
20	Sales	Sales department	ACCESS1 Fa0/2; ACCESS2 Fa0/2	Po1, Po2 and Po3 trunks
30	IT	IT department	ACCESS1 Fa0/3; ACCESS2 Fa0/3	Po1, Po2 and Po3 trunks

The names Management, Sales and IT are derived from the access-port descriptions LAN_Mgmt, LAN_Sales and LAN_IT. The running-config exports do not show separate vlan name commands, so these names are treated as functional project labels.

7.1. Traffic paths by VLAN

- VLAN 10 traffic between the two locations crosses Po2, Po1 and Po3, depending on source and destination.
- VLAN 20 uses the same trunk path while remaining isolated from the Management and IT broadcast domains.
- VLAN 30 should use the same end-to-end path, but the inconsistent STP Root ID must be resolved first.
- Traffic between VLANs is not possible without a router or multilayer SVI interfaces.

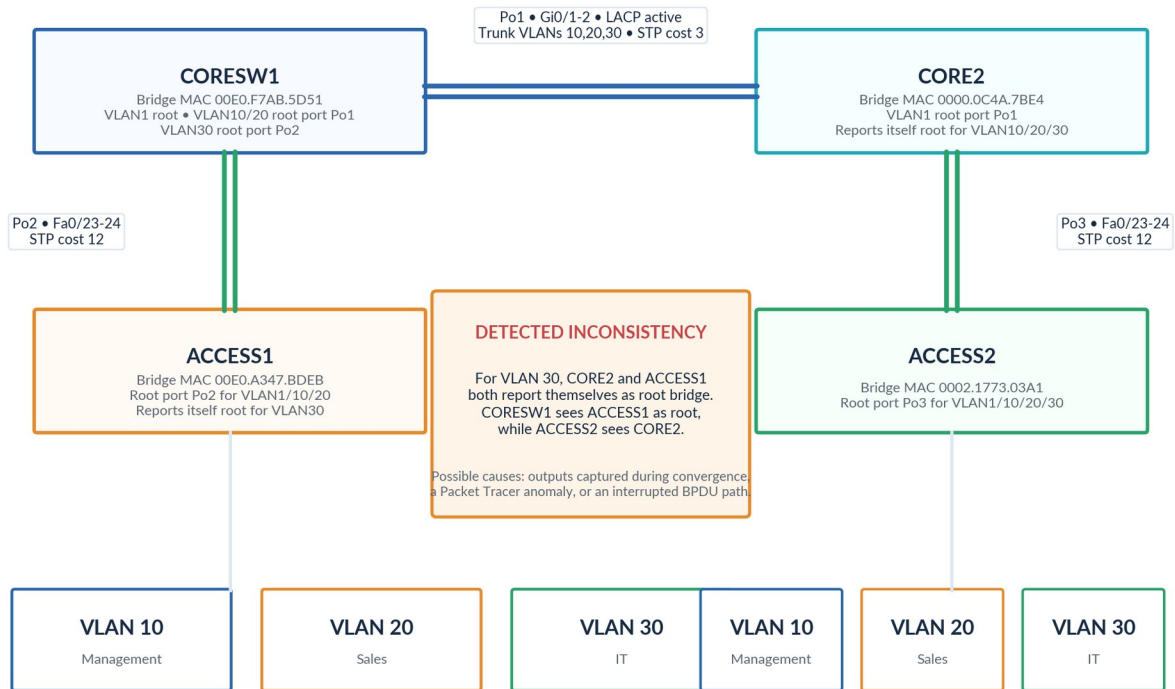
LAYER 3 LIMITATION

VLAN segmentation alone does not provide communication between departments. The current configuration intentionally demonstrates Layer 2 functions only.

8. Detailed network diagram

Detailed L2 diagram: EtherChannel, trunks and STP roles

The state is reconstructed from four separate show spanning-tree outputs.



Target state: every device must display the same Root ID for each VLAN; priorities should be explicitly configured for VLANs 10, 20 and 30.

Figure 2 - EtherChannel links, STP roles and the inconsistent VLAN 30 state

The diagram shows logical port-channel links instead of six independent physical links. Po1 has an STP cost of 3, while Po2 and Po3 have a cost of 12. Because every physical link pair is successfully bundled, STP should treat each EtherChannel as one logical port.

9. Spanning Tree Protocol analysis

PVST is enabled on all devices. Only VLAN 1 has manually configured priorities: CORESW1 uses 24576 and CORE2 uses 28672. VLANs 10, 20 and 30 use the default priority of 32768, so the root bridge is selected by the lowest MAC address when priorities are equal.

VLAN	Reported Root ID	Root ports	Consistent	Interpretation
1	CORESW1 / 00E0.F7AB.5D51	CORE2 Po1; ACCESS1 Po2; ACCESS2 Po3	Yes	Effective root priority 24577
10	CORE2 / 0000.0C4A.7BE4	CORESW1 Po1; ACCESS1 Po2; ACCESS2 Po3	Yes	CORE2 wins through the lower MAC address
20	CORE2 / 0000.0C4A.7BE4	CORESW1 Po1; ACCESS1 Po2; ACCESS2 Po3	Yes	Same state as VLAN 10
30	Split: ACCESS1 or CORE2	CORESW1 Po2; ACCESS2 Po3; local root on A1 and C2	No	The four outputs cannot represent one converged topology

9.1. Why are no EtherChannel ports blocked?

After aggregation, the physical members of a channel represent one logical STP port. The logical topology CORESW1-CORE2-ACCESS2 plus CORESW1-ACCESS1 contains no closed loop across different port-channels, so STP has no alternate link that it must block. Redundancy exists inside each channel, not through a second network path.

10. EtherChannel and trunk analysis

Channel	Link	Members	Negotiation	Operational state	Trunk/STP
Po1	CORESW1 ↔ CORE2	Gi0/1 + Gi0/2	LACP active / active	SU; both members P	Trunk VLANs 10,20,30; cost 3
Po2	CORESW1 ↔ ACCESS1	Fa0/23 + Fa0/24	LACP active / active	SU; both members P	Trunk VLANs 10,20,30; cost 12
Po3	CORE2 ↔ ACCESS2	Fa0/23 + Fa0/24	LACP active / active	SU; both members P	Trunk VLANs 10,20,30; cost 12

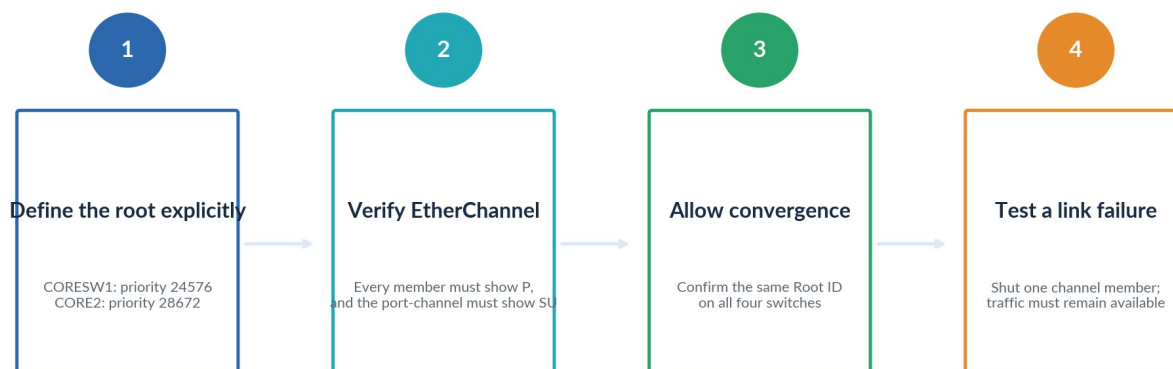
The SU flag means the port-channel is a Layer 2 channel that is in use, while P beside each member means that the interface is correctly bundled into the aggregator. The trunk status is trunking, encapsulation is 802.1Q, the native VLAN is 1, and the allowed and active VLANs are 10, 20 and 30.

10.1. Expected failure behavior

- Failure of one physical member: the port-channel remains active through the remaining member, with lower aggregate capacity.
- Failure of both Po2 members or CORESW1: ACCESS1 loses connectivity to the rest of the network.
- Failure of both Po3 members or CORE2: ACCESS2 loses connectivity to the rest of the network.
- Failure of both Po1 members: the network splits into two isolated core/access domains.

11. Problem and solution - inconsistent STP topology

Problem and recommended resolution process



Apply the commands in a controlled sequence, then repeat show spanning-tree vlan 10, 20 and 30 on every switch.

Figure 3 - Recommended process for stabilizing the STP state

11.1. Detected problem

For VLAN 30, the supplied show spanning-tree outputs report two different root bridges. CORESW1 sees ACCESS1 as root and uses Po2 as its root port; ACCESS1 reports itself as root. At the same time, CORE2 reports itself as root and ACCESS2 sees CORE2 as root through Po3. These two views cannot simultaneously exist in one fully converged PVST instance if Po1 is truly carrying VLAN 30.

11.2. Most likely causes

- The outputs were captured at different times, before STP convergence completed.
- Priorities for VLANs 10, 20 and 30 are not explicitly defined, so root election depends on MAC addresses.
- VLAN 30 or its BPDUs temporarily did not cross Po1 even though the later trunk output shows the VLAN as active.
- Cisco Packet Tracer displayed stale or inconsistent STP state; the core outputs show the port-channel and some physical members at the same time.

11.3. Recommended solution

The root bridge should be defined explicitly at the core layer for all three user VLANs. The simplest plan is to make CORESW1 the primary root and CORE2 the secondary root for VLANs 10, 20 and 30. This removes MAC-based election and gives every device the same expected Root ID.

```
! CORESW1
configure terminal
spanning-tree vlan 10,20,30 priority 24576
end

! CORE2
configure terminal
spanning-tree vlan 10,20,30 priority 28672
end
```

After applying the configuration, verify that VLAN 30 exists and is allowed on both sides of Po1, allow at least one STP convergence interval, and repeat verification on all four devices.

```
show vlan brief
show interfaces trunk
show etherchannel summary
show spanning-tree vlan 10
show spanning-tree vlan 20
show spanning-tree vlan 30
```

SUCCESS CRITERION

For each VLAN, all four switches must display the same Root ID. Po1, Po2 and Po3 must show SU, and every member must show P. If Packet Tracer continues to display stale state, Po1 may be administratively restarted during a controlled lab window, or the project may be reopened, followed by another verification.

12. Security and management

Control	Status	Effect/note
SSHv2	Enabled on all four switches	VTY permits only SSH; Telnet is not allowed.
Local AAA	Enabled	Login uses the local database; confidential hash values are omitted from this document.
PortFast	Fa0/1-3 on both access switches	Accelerates endpoint-port activation.
BPDU Guard	Fa0/1-3 on both access switches	Protects against an additional switch connected to a user port.
Unused-port shutdown	Fa0/4-22 on access and Fa0/1-22 on core	Reduces the opportunity for unauthorized connections.
DTP nonegotiate	Po1, Po2 and Po3	Prevents dynamic trunk negotiation.
Management IP	Not configured	SSHv2 is not remotely reachable without an SVI address and Layer 3 path.
Native VLAN 1	Default	Recommendation: use a dedicated unused native VLAN in production.

13. Test and verification plan

#	Test	Command/procedure	Expected result
1	EtherChannel state	show etherchannel summary	Po1/Po2/Po3 = SU; every member = P
2	Trunk state	show interfaces trunk	Port-channels are trunking; allowed/active VLANs 10,20,30
3	STP root consistency	show spanning-tree vlan 10,20,30	Same Root ID on all four switches for each VLAN
4	VLAN assignment	show vlan brief	Fa0/1 in VLAN10, Fa0/2 in VLAN20, Fa0/3 in VLAN30
5	Single-member failure	Shut down one channel member	Port-channel remains up over the other member
6	Member recovery	no shutdown	Member returns to P without changing the logical topology
7	BPDU Guard	Controlled BPDU test on an access port	Port becomes err-disabled, followed by safe recovery
8	Layer 2 reachability	Ping within one VLAN using known host IPs	Same VLAN works; inter-VLAN traffic fails without a router/SVI

14. Limitations and recommendations

Limitation/finding	Recommendation	Priority
STP priority only for VLAN 1	Explicitly configure primary/secondary root roles for VLANs 10,20,30.	High
Inconsistent VLAN 30 Root ID	Check VLAN/trunk state, allow convergence and repeat show outputs.	High
No management addressing	Add a dedicated management VLAN/SVI, IP plan and default gateway only if Layer 3 management is a lab objective.	Medium
No inter-VLAN routing	Add multilayer SVIs or router-on-a-stick if communication between departments is required.	As needed
Access switches are not dual-homed	Connect each access device to both core switches and design STP/EtherChannel with a supported technology.	Medium
Access Gi0/1-2 are not disabled	Administratively shut them down if they are unused.	Medium
Native VLAN 1	Use a dedicated unused native VLAN and match both trunk ends.	Medium
Exact model/host addresses not confirmed	Export show version, show inventory and document endpoint IP settings from the .pkt project.	Low

15. Conclusion

The project is a well-structured Layer 2 lab that successfully demonstrates VLAN segmentation, static 802.1Q trunks and LACP EtherChannel. All three port-channels are operational: the channels are Layer 2 and in use, and all physical members are correctly bundled. Access ports are consistently assigned to VLANs 10, 20 and 30 and protected by PortFast and BPDU Guard.

The main technical deficiency is the non-deterministic and inconsistent STP state for VLAN 30. Because user-VLAN priorities remain at their default value, root election depends on MAC addresses, and the four supplied outputs show two separate views of VLAN 30. Explicitly defining CORESW1 as the primary root and CORE2 as the secondary root, followed by trunk verification and renewed convergence, is the most important correction.

FINAL ASSESSMENT

The network is functional for demonstrating Layer 2 technologies and link-level redundancy. After stabilizing the STP root plan, shutting down the remaining unused ports and optionally adding management addressing, the project provides a clear and solid foundation for more advanced campus labs.

Appendix A - Sources and key commands

A.1. Supplied sources

- Lab04_STPthChannel.pkt - Cisco Packet Tracer project.
- SWCORE 1.txt - CORESW1 running-config, STP, EtherChannel and trunk outputs.
- SWCORE2.txt - CORE2 running-config, STP, EtherChannel and trunk outputs.
- ACCESS 1(1).txt - ACCESS1 running-config, STP, EtherChannel and trunk outputs.
- ACCESS 2(1).txt - ACCESS2 running-config, STP, EtherChannel and trunk outputs.

A.2. Key configuration blocks

```
spanning-tree mode pvst

interface Port-channel1
 switchport trunk allowed vlan 10,20,30
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport nonegotiate

interface FastEthernet0/1
 switchport access vlan 10
 switchport mode access
 spanning-tree portfast
 spanning-tree bpduguard enable

interface FastEthernet0/23
 channel-group <2-or-3> mode active
```

Note: usernames, enable secrets and other authentication hash values are intentionally not reproduced in this documentation.